Chapter 6 — Data types and central tendency

Before plotting, classify each column: continuous versus discrete numerics, nominal versus ordinal categories

Learning objectives

- Distinguish four common data types, compute mean, median
- Mode
- State when each central tendency measure is appropriate

Types of data

- Continuous variables take any value in an interval, height, income, sales totals
- Discrete variables are counts, purchases per month, items in a basket
- Nominal categories have no order, product type, color
- Ordinal categories have meaningful order, satisfaction ratings, education bands

Example 6.10 and 6.13 — Continuous and nominal cases

- Example 6.10 — hands-on module
- Example 6.10 lists continuous physical measures
- Example 6.13 shows nominal product categories
- Explore the chapter example module
- View files: `modules/chapter6/example10/`

Mean

- The mean is the arithmetic average, sum divided by count
- Example 6.15 computes mean income
- Use means for roughly symmetric continuous data

Median

- The median is the middle value when sorted
- Example 6.16 contrasts odd and even samples
- Median resists outliers and suits skewed income or spend distributions

Mode

- The mode is the most frequent value, especially useful for categorical data
- Example 6.17 shows a simple numeric mode
- A multimodal distribution may indicate subpopulations worth splitting

Matching summary to type

- Use means and medians for numeric exploration; use mode or frequency tables for categories
- Report all three for numeric fields during early EDA so skew is visible when mean and

Takeaways

- Variable type dictates valid summaries
- Mean, median, and mode answer different questions about center
- Always inspect type before computing describe() output blindly

Next

- Complete the quiz for this part
- Continue to dispersion, shape, frequency tables, and grouping